

Jacob Edwards, PhD

Neurophysiologist & Data Scientist

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Scientist with 13 years' experience using data and code to draw insights from complex, dynamic systems. Published 6 peer-reviewed data papers cited 80+ times, secured \$100k+ in research funding, and created and taught highly-rated college courses in statistics and applied data science. Specialized in time-series signals including biosensor data and acoustic recordings like speech and music. Delivers meticulously crafted figures and reports that turn complex quantitative results into clear and impactful visual stories.

SKILLS

Tools

Python, R, MATLAB, SQL, Git, MS Office

Data Analysis

Statistical modeling, regression and classification, time series analysis, natural language processing, machine learning, feature engineering, dimensionality reduction, model evaluation and prediction, data cleaning, wrangling, and visualization

Experimentation

Hypothesis testing, experimental design, sampling methods, effect size estimation, bootstrapping, simulation

Communication

Data storytelling, technical writing, oral presentation, cross-functional collaboration, teaching and mentoring

RELEVANT EXPERIENCE

Research Scientist	COLUMBIA UNIVERSITY	New York, NY	2018 - 2025
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Time-Series Modeling of Brain Data

- Generated and analyzed 3.2 TB of time-series neural data. Built scalable workflows for data cleaning, feature extraction, aggregation, and visualization.
- Implemented methods to understand and predict stimulus-response relationships, including kernel smoothing, adaptive thresholding, auto- and cross-correlation functions, Fourier transforms, and receptive field estimation.
- Automated a data quality screening workflow by training and deploying a random forest [anomaly detection classifier](#), reducing manual review time by ~20%.
- Built custom [interactive dashboards](#) to support exploratory analysis and enable non-technical collaborators to interpret multi-dimensional results.

Sequence Modeling of Vocal Acoustic Data

- Generated and analyzed a dataset of 30,000+ vocal acoustic events from 179 subjects. Optimized event segmentation pipeline to derive structured tabular data from raw audio waveforms.
- Unified infrastructure for database management and annotation by creating a [graphical user interface](#) that reduced work hours per project by ~80% and was used internally by 3 other teams.
- Engineered features using domain metrics and decomposition methods (PCA, wavelets) for downstream cluster analysis to identify latent acoustic structure.
- Developed and evaluated NLP model for sequence prediction using non-lexical signal features. Designed custom cost metric based on fuzzy matching algorithms. Created custom effect size metrics and bootstrapped confidence intervals.

Leadership and Cross-Functional Impact

- Secured \$140k in competitive [research funding](#) by articulating project proposals as feasible aims and measurable outcomes.
- Created and taught a new [applied data science curriculum](#) (2 years taught, 20 hours of lecture, 7 homework assignments) covering R programming, data wrangling and visualization, predictive analytics, machine learning, and more. Emphasized translating real-world data analysis into actionable policy recommendations for legislators and decision-makers.
- Delivered presentations to technical and non-technical audiences internally, at national conferences, and at community events including *Taste of Science NYC* and *BioBus Harlem*.
- Authored 6 peer-reviewed papers in high-impact journals including *Nature Scientific Reports* and *Journal of Neuroscience*.

SELECT PEER-REVIEWED PUBLICATIONS

2026	Models of species' syntax rules accurately predict vocal sequences	[paper]
2025	Temporal organization of behavior is driven by species identity	[paper]
2023	Machine learning analysis of song links vocal acoustics to ancestry	[paper]

EDUCATION

PhD	2025	Columbia University
MSc	2018	Colorado State University
BSc	2013	Uni. of Tennessee, Knoxville